

LAPORAN PRATIKUM IoT II
MODUL 3



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PENULIS :

Nama : M Haikal Pasya

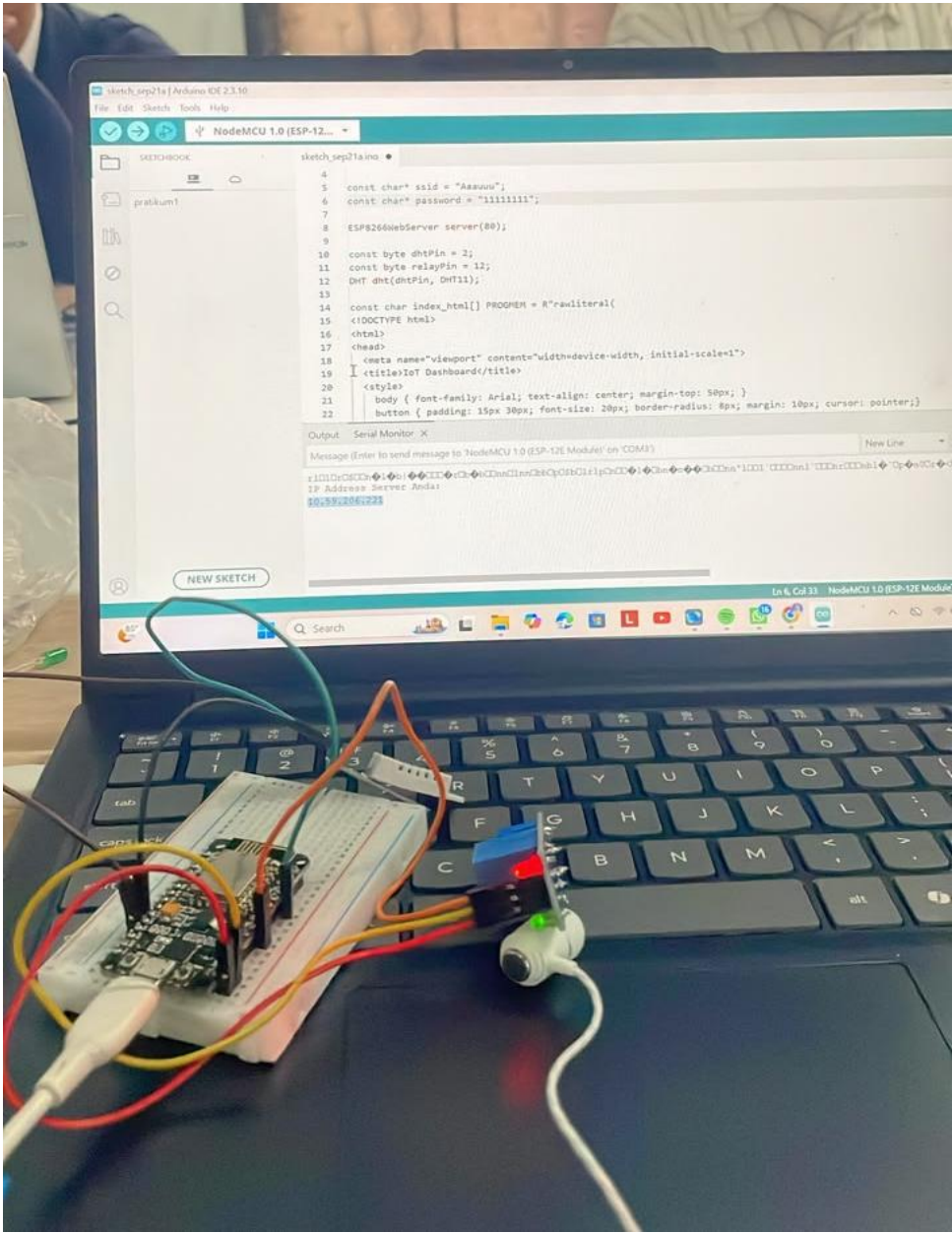
Kelas : TRI 3B

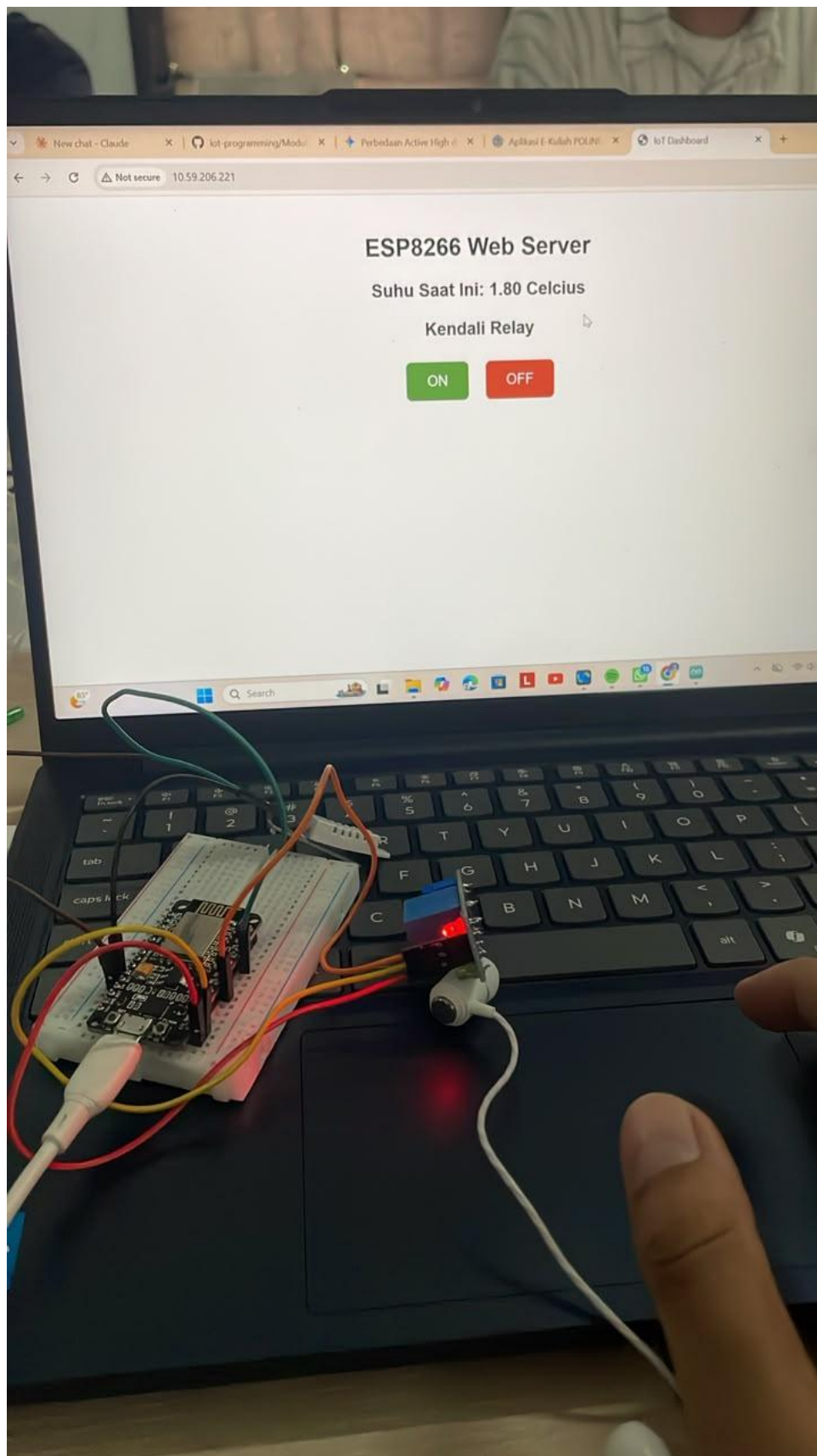
Npm : 25782043

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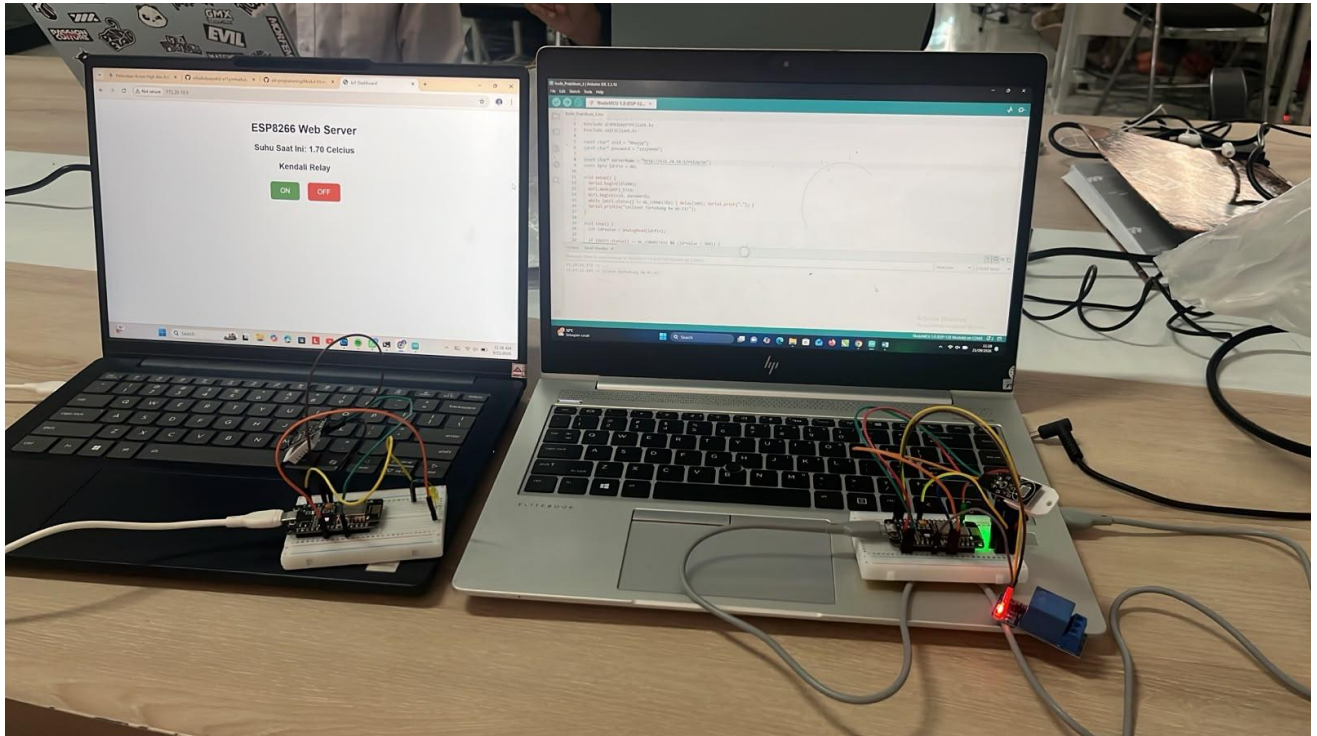
PRATIKUM 1



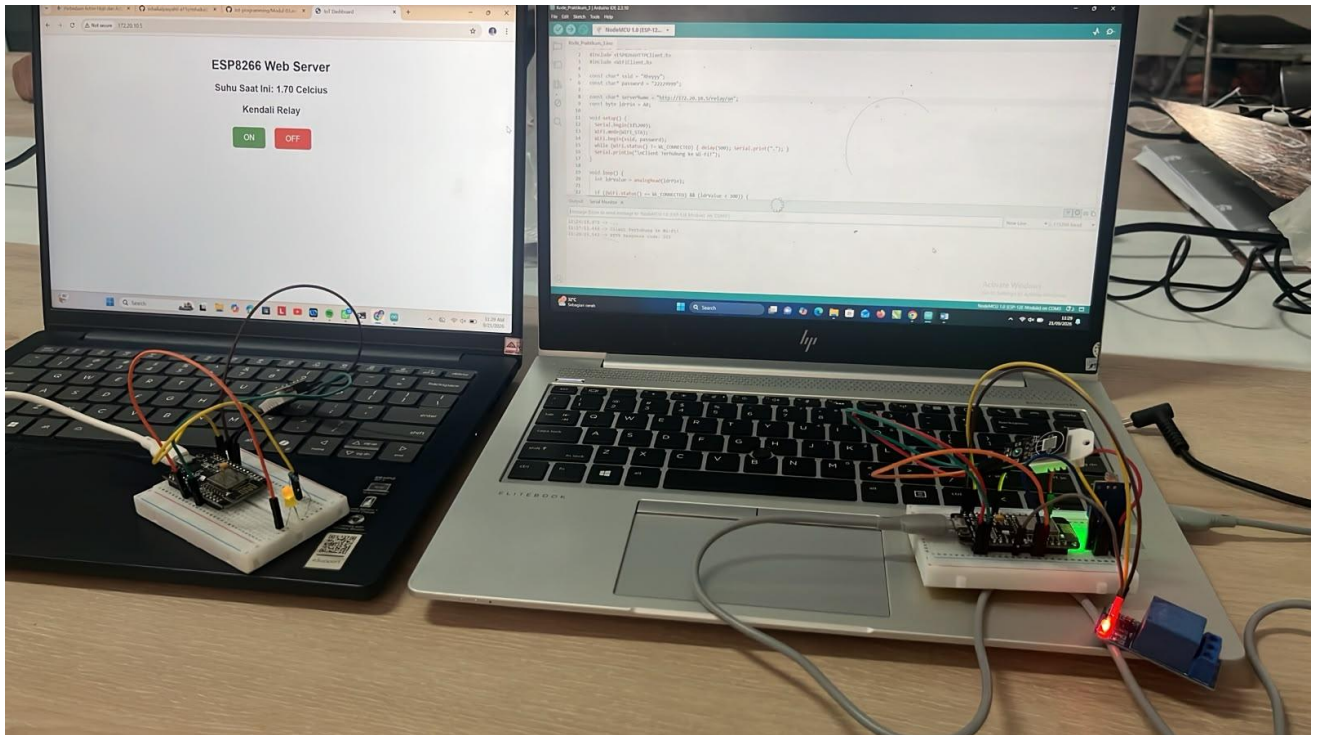


PRATIUM 2:

LED HIDUP:



LED MATI:

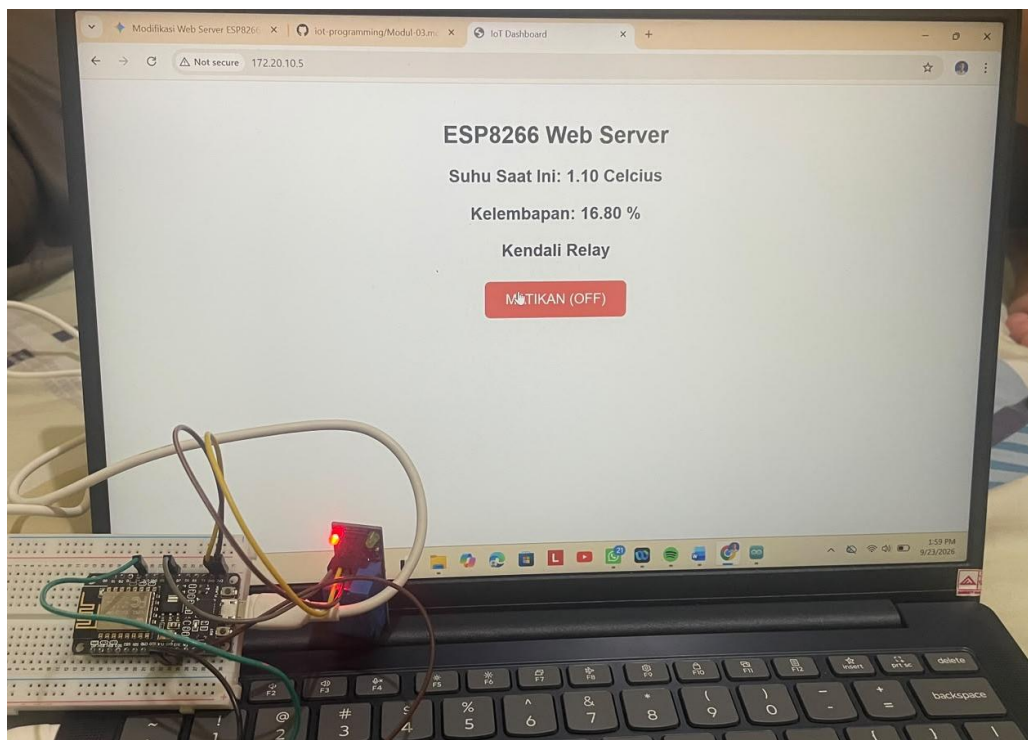


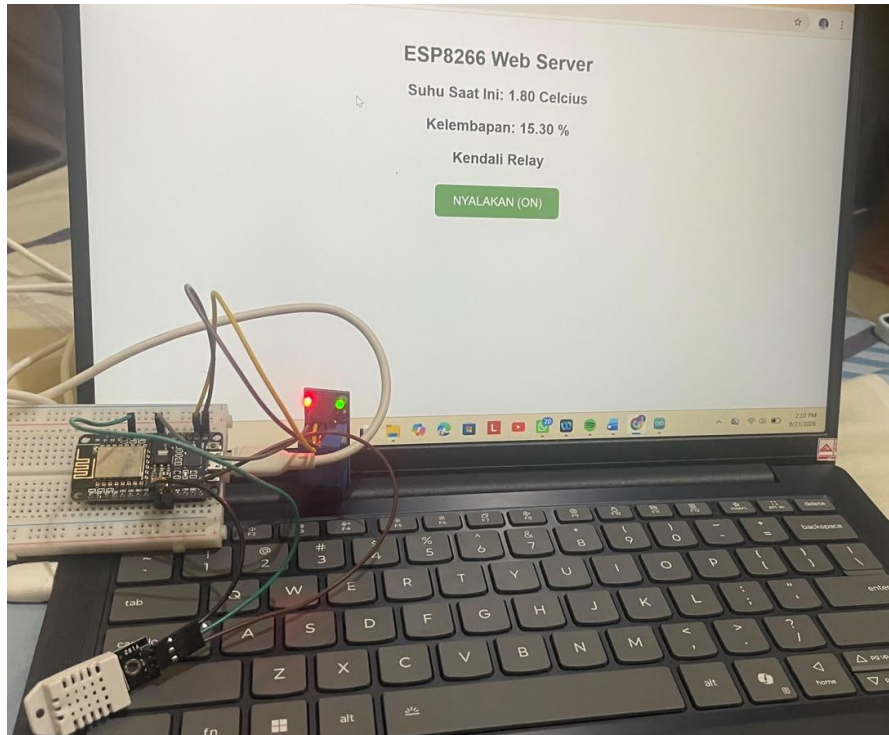
LATIHAN :

LATIHAN 1 : Berdasarkan wawasan dari subbab 3.3.1, sebutkan secara logis mengapa kita Menggunakan mode jaringan Station (STA) pada kedua pratikum diatas, dan bukan mode Access Point (AP)?

JAWAB: Kita menggunakan mode jaringan Station (STA) dan bukan mode *Access Point (AP)* karena dalam mode STA, ESP8266 bertindak sebagai klien yang terhubung ke jaringan router atau *hotspot* yang sudah ada (yang memiliki koneksi internet atau terhubung ke jaringan lokal utama).

LATIHAN 2: (Modifikasi Antarmuka & Telemetry):



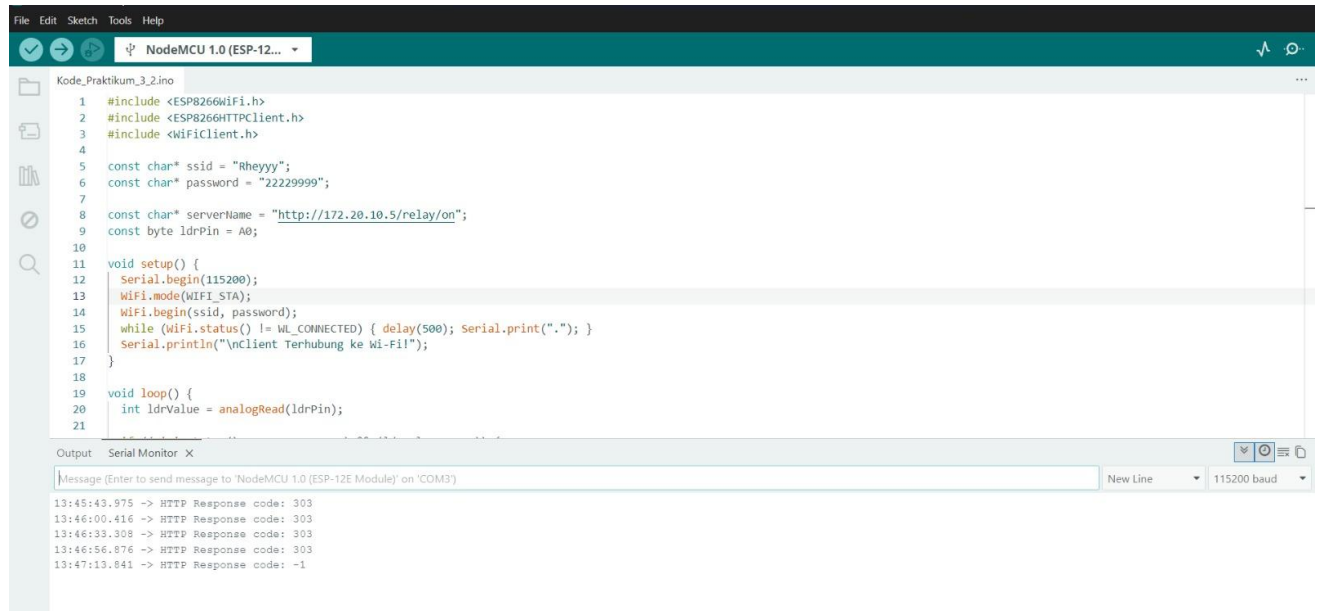


```

14  const char index_html[] PROGMEM = R"rawliteral(
15  <!DOCTYPE html>
16  <html>
17  <head>
18      <meta name="viewport" content="width=device-width, initial-scale=1">
19      <title>IoT Dashboard</title>
20      <style>
21          body { font-family: Arial; text-align: center; margin-top: 50px; }
22          button { padding: 15px 30px; font-size: 20px; border-radius: 8px; margin: 10px; cursor: pointer; }
23          .btn-on { background-color: #4CAF50; color: white; border: none; }
24          .btn-off { background-color: #f44336; color: white; border: none; }
25          .sensor-box { font-size: 24px; font-weight: bold; }
26      </style>
27  </head>
28  <body>
29      <h1>ESP8266 Web Server</h1>
30      <div class="sensor-box">
31          <p>Suhu Saat Ini: <strong>%TEMPERATURE%</strong> Celcius</p>
32          <p>Kelembapan: <strong>%HUMIDITY%</strong> %</p>
33      </div>
34  </body>
35  </html>
36  )rawliteral";
37
38  void handleRoot() {
39      String html = index_html;
40      float t = dht.readTemperature();
41      float h = dht.readHumidity();
42
43      if (isnan(t)) {
44          html.replace("%TEMPERATURE%", "--");
45      } else {
46          html.replace("%TEMPERATURE%", String(t));
47      }
48
49      // Mengganti teks placeholder %HUMIDITY% dengan kelembapan nyata[cite: 3]
50      if (isnan(h)) {
51          html.replace("%HUMIDITY%", "--");
52      } else {
53          html.replace("%HUMIDITY%", String(h));
54      }
55  }

```

LATIHAN 3 : (Analisis Kode Client): Pada Praktikum 2, *Serial Monitor* Client mencetak HTTP Response code. Analisislah apa yang akan dicetak oleh *Serial Monitor* Anda jika ternyata NodeMCU Server teman Anda mendadak kehilangan daya (*offline*)? (Silakan cabut daya Server dan lihat hasil erornya).

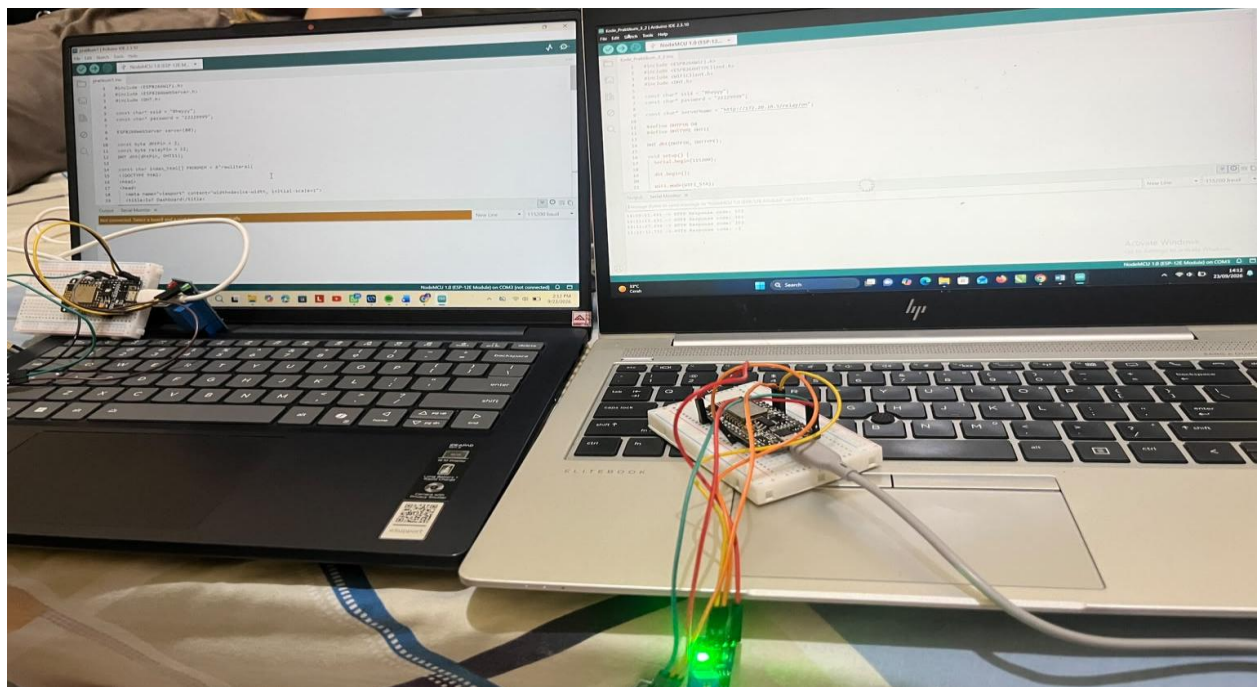


The screenshot shows the Arduino IDE interface. The code editor displays the following code for `Kode_Praktikum_3_2.ino`:

```
1 #include <ESP8266WiFi.h>
2 #include <ESP8266HTTPClient.h>
3 #include <WiFiClient.h>
4
5 const char* ssid = "Rheyyy";
6 const char* password = "22229999";
7
8 const char* serverName = "http://172.20.10.5/relay/on";
9 const byte ldrPin = A0;
10
11 void setup() {
12   Serial.begin(115200);
13   WiFi.mode(WIFI_STA);
14   WiFi.begin(ssid, password);
15   while (WiFi.status() != WL_CONNECTED) { delay(500); Serial.print("."); }
16   Serial.println("\nClient Terhubung ke Wi-Fi!");
17 }
18
19 void loop() {
20   int ldrValue = analogRead(ldrPin);
21 }
```

The Serial Monitor window shows the following output:

```
13:45:43.975 -> HTTP Response code: 303
13:46:00.416 -> HTTP Response code: 303
13:46:33.308 -> HTTP Response code: 303
13:46:56.876 -> HTTP Response code: 303
13:47:13.841 -> HTTP Response code: -1
```



Latihan 4 :

(Pengembangan Algoritma): **Anda ingin mendesain Klien Pendeteksi Suhu.**

Modifikasi *conditional block* if di Praktikum 2 sehingga *Client* mengirim permintaan (GET Request) HANYA JIKA sensor DHT membaca suhu melebihi 35 Celcius (bukan menggunakan LDR). Tulis baris algoritmanya.

```
32 void loop() {
33     float temperature = dht.readTemperature();
34
35     Serial.print("Suhu: ");
36     Serial.print(temperature);
37     Serial.println(" C");
38
39     if ((WiFi.status() == WL_CONNECTED) && (temperature > 35)) {
40         WiFiClient client;
41         HTTPClient http;
42
43         http.begin(client, serverName);
44
45         int httpResponseCode = http.GET();
46
47         Serial.print("HTTP Response code: ");
48         Serial.println(httpResponseCode);
49
50         http.end();
51
52         delay(10000);
53     }
54
55     delay(2000);
56 }
```

Tugas : **Pengembangan *Web Server* Cerdas dan Analisis Arsitektur**

